

Behavioral Decision-Making Approach for Vehicle Platoon Control: Two Noncooperative Game Models

Yang Liu^{ID}, Changfu Zong, Changhua Dai^{ID}, Hongyu Zheng^{ID}, and Dong Zhang^{ID}, *Member, IEEE*

Abstract—Game theory has been extensively studied in recent years for its potential benefits on solving interactions between autonomous vehicles in platoon control. In this article, we propose two behavioral decision-making approach based on non-cooperative game theory with both the complete information and incomplete information for cooperative vehicle platoon systems. The non-cooperative game payoff function takes the platooning performances of economy, comfort, safety, and further achieving self-driving functions into consideration. For the driving situations with incomplete information, a belief pool is constructed to represent the action probability for different behavioral types of vehicles, which will be updated by combining the driving intention identification with a Bayesian probability formula. With this, the stable strategies can be obtained for the two potentially conflicting parties, ensuring that neither of them has a motivation to change their driving behavior. Finally, the simulation results demonstrate that with the proposed behavioral decision-making approaches with complete information, the cooperative platoon performance can be markedly improved; with incomplete information, the platoon can determine the behavioral types of conflicting vehicles and complete collaborative decision-making approaches for solving not only the simple road-rights conflicting problems, but also for the extended vehicle platoon driving scenarios in a more complex transportation system.

Index Terms—Behavioral decision, game theory, Nash equilibrium, vehicle platoon.

I. INTRODUCTION

COOPERATIVE control of a vehicle platoon allows multiple vehicles to travel in a specific formation by automatically adjusting the distance and yaw angle between them. When traveling appreciable distances, such formations effectively reduce peak acceleration and avoid excessive acceleration and deceleration operations, thereby saving energy, reducing pollution, and improving traffic efficiency. Hence, due to the potential advantages listed above, research into vehicle platoons has significantly increased [1], [2], [3].

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Yang Liu, Changfu Zong, Changhua Dai, and Hongyu Zheng are with the State Key Laboratory of Automotive Simulation and Control, Jilin University, Changchun 130022, China (e-mail: lyang17@mails.jlu.edu.cn; zongcf@jlu.edu.cn; zhenghy@jlu.edu.cn).

Dong Zhang is with the Department of Mechanical and Aerospace Engineering, Brunel University London, UB8 3PH Uxbridge, U.K. (e-mail: dong.zhang@brunel.ac.uk).

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Vehicle platoon studies can be classified as 1-D or 2-D, according to the number of degrees of freedom for vehicle movement. 1-D platoon coordination mainly focuses on spacing control strategies, acceleration-decision methods, and string stability [4], [5], [6], while road-rights solution and traffic efficiency are more important research topics in 2-D scenarios. The former topic concerns the resolution of spatial position conflicts for any number of vehicles at any moment in time, thereby ensuring safety, while the emphasis of the latter topic is to reduce the longitudinal speed fluctuations caused by changes in the lateral position of the vehicle as much as possible, which cause the problem of “Phantom Traffic Jams.” The road-rights solution can be abstracted as the mathematical problem of behavioral decision-making for obtaining the optimal traffic sequence. Uncontrolled intersections, cut-in of traffic in the ramp area, and multi-vehicle collaborative lane change situations are the three main traffic-flow scenarios that have been identified as requiring cooperative behavioral decision-making. The behavioral decision-making methods adopted by road-rights conflict solution techniques above are categorized as “ad hoc negotiation based” strategies [8], [9], [10], [11], and “global planning based” strategies [12], [13], [14], [15], [16]. While ad hoc negotiation based strategies are more practical to implement, special situations in which an interacting vehicle cannot be controlled are not considered. The controlled vehicle makes behavior decisions according to preset rules lacks flexibility, and it is impossible to fully consider the demand characteristics of all vehicles. As the number of participating vehicles increases, the computational complexity for conflict resolution sharply rises. Thus, in conflict scenarios, a behavioral decision-making method that considers the characteristics of different vehicle-types, as well as the cooperation characteristics of the vehicle platoon, and competition between the vehicle platoon and external vehicles, has important research significance.

In recent years, scholars have explored the application of game theory concepts to behavioral decision-making methods for vehicles [17], [18]. For example, Basiri et al. [19] used game theory to solve the security problem of the information flow topology of vehicle platoons, adopting a Nash equilibrium to select the appropriate security improvement strategy; Gattami et al. [20] used optimal control and game theory to determine the minimum relative distance required to ensure safety, given the conflict between safety and fuel consumption in the platoon; and Farokhi and Johansson [21] used a congestion game to explore the motivation of drivers to form a

platoon. While these studies have focused on game theory in cooperative platoon control, others have concerned the conflict between vehicles in lane-changing. For instance, Yu et al. [22] proposed a novel game-theory-based model, using interactions with surrounding drivers based on turn signals and lateral movement to simulate the lane-changing behavior of human drivers. Similarly, Elhenawy et al. [23] proposed a game-theory-based algorithm for controlling autonomous vehicles at uncontrolled intersections, with simulations demonstrating that this technique could reduce travel time and delay. Ding et al. [24] proposed a coordinated multivehicle strategy for mandatory lane changes that improved the average speed of the vehicles and driving stability in simulations, while Kang and Rakha [25] developed a repeat-game-theoretical decision-making model with an updated payoff function for human drivers.

With the above, we can conclude that, cooperative behavioral decision-making between platoon and other vehicles outside the platoon can be considered as the interaction of multi-agents, where one's action will affect other's decision-making result. Vehicles in this interaction prefer to take the actions to benefit themselves. Therefore, game theory which has not been used to coordinate platoon-level behavior decisions in scenarios where the platoon conflicts with other vehicles, is suitable for solving relative interaction. In addition, if there is no unified payoff function standard, or there are communication protocol restrictions, as in actual driving scenarios, conflicting vehicles may not be able to fully obtain each other's payoffs for a complete information game. Unlike the methods proposed in other studies, including reinforcement learning, Bayesian optimization which can be used for similar issues, the objective of using game theory is to find an equilibrium solution rather than an optimal solution [26], [27]. The equilibrium solution can ensure that the participants in the traffic flow are not willing to actively change their actions to remain a stable state.

To address this, we abstract road-rights conflict scenarios with complete information and incomplete information as noncooperative game model. In complete information game model, Nash equilibrium is used to compare different interactive mode, including cooperative mode and selfish mode. In incomplete information game model, our technique uses a shared belief pool for collaborative decision-making inside the platoon. This is then combined with the Bayesian formula, to obtain the posterior probability for other interacting party types, enabling a more accurate selection of the behavior strategy with the highest expected payoff.

Note that, the approaches proposed in this article demonstrate the potential benefits for a more complex vehicle platoon driving scenarios rather than for solving the simple road-rights conflicting problems.

The innovative point in this approach comes from the idea that conducting road-right game between the vehicles inside the platoon and non-cooperative game between the external vehicles and the whole vehicle platoon. The feasibility of theoretical application of cooperative game and noncooperative game in this scenario is explored. The main theoretical contributions of this study are summarized as follows.

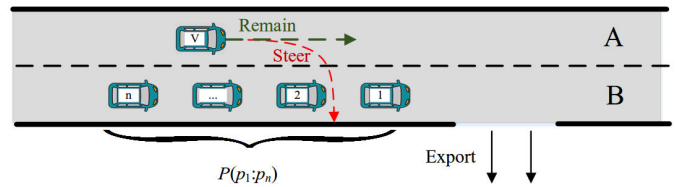


Fig. 1. Schematic illustration of the cut-in scenario.

- 1) This article proposed a complete theoretical system based on non-cooperative game theory with both complete information and incomplete information for the cooperative vehicle platoon systems. It systematically analyzes all the potential driving conditions which the platoon might face in the platoon-level behaviors, and proposed a series of feasible solutions aiming at the strategical decision-making level.
- 2) Two cost function which indicate cooperative interactive mode and selfish interactive mode are presented. After deriving the equilibrium existence conditions, the solutions of two cost functions are obtained to compare two different interactive modes.
- 3) To address typical scenarios where a vehicle cannot obtain the other's precise payoff function, we devise a payoff function that features a belief pool for estimating different behaviors. The conditions for obtaining the equilibrium solution for the incomplete information dynamic game modeling in the described conflict scenario are also explored.

The remainder of this article is outlined as follows. The cut-in scenario and behavioral decision-making framework are introduced in Section II, where the non-cooperative platoon game is defined. An acceleration decisional method is proposed in Section III for conflict situations with complete information. In Section IV, a dynamics model with incomplete information is presented, in addition to the analysis of the refined Bayesian Nash equilibrium used to obtain the balanced strategy set. Simulations of our technique and results are presented in Section V, with conclusion discussed in Section VI.

II. PROBLEM FORMULATION

The problem of platoon-vehicle interaction is defined as in Fig. 1. In the cut-in scene depicted in this image, it is assumed that platoon P ($P(p_1: p_n)$) of length d is driving along Lane B. Vehicle V's driving plan requires it to move from Lane A to Lane B, and subsequently exit from the current road. At the time depicted in the image, the platoon occupies the lane space, making Vehicle V unable to change lanes freely. Some vehicles in P have detected the presence of Vehicle V in the adjacent lane, which can detect conflicting vehicle status information when changing lanes.

In the situation detailed above, the participants are divided into two categories: a single-vehicle outside the platoon and vehicles inside the platoon. It is assumed that all vehicles aim to maximize payoff, and the strategies adopted by both parties will affect their payoff, including their driving safety, economy, and completion of driving intention. Therefore, this problem can be abstracted as a game problem.

The non-cooperative platoon game consists of vehicles outside the platoon, designated as $V = \{v_1, v_2, \dots, v_i\}$, and the vehicles in the platoon, $P = \{p_1, p_2, \dots, p_j\}$. This game can be written as $G(V, P, v_v, v_p, \sigma_v, \sigma_p)$, where $v_v = \{v_v^1, v_v^2, \dots, v_v^i\}$ denotes the payoff vector of the vehicles outside the platoon, $v_p = \{v_p^1, v_p^2, \dots, v_p^j\}$ denotes the payoff vector of vehicles in the platoon, $\sigma_v = \{\sigma_v^1, \sigma_v^2, \dots, \sigma_v^i\}$ represents the action set of vehicles outside the platoon, and $\sigma_p = \{\sigma_p^1, \sigma_p^2, \dots, \sigma_p^j\}$ is the action set of vehicles in the platoon. Each action set and payoff vector has a unique mapping, which is the payoff function of the non-cooperative platoon game.

To define the payoff functions of interacting parties, we assume that the action set for single vehicles is {steer, remain}. Here, “steer” means that the single-vehicle drives to another lane, while “remain” denotes that it maintains its position in its original lane. In response, the action set of vehicles in the platoon is {permit, refuse}, i.e., they can either allow or deny single vehicles access to the lane they require. The platoon’s payoff is given as $v_p(\sigma_p, \sigma_v)$ and the single vehicle’s payoff is given as $v_v(\sigma_v, \sigma_p)$. During the game, the payoff function needs to account for all aspects of the vehicle’s performance while driving. Hence, it includes expressions for an economic payoff, safety payoff, and driving function payoff. The overall payoff functions for a single vehicle and the platoon are expressed as below

$$\begin{cases} v_v = \alpha_v v_{\text{eco}} + \beta_v v_{\text{safe}} + \gamma_v v_{\text{fun}} \\ v_p = \alpha_p v_{\text{eco}} + \beta_p v_{\text{safe}} + \gamma_p v_{\text{fun}} \end{cases} \quad (1)$$

where α_v, β_v , and γ_v denote weight coefficients for the single vehicle’s payoff function, α_p, β_p , and γ_p denote weight coefficients for the platoon’s payoff function, and $v_{\text{eco}}, v_{\text{safe}}$, and v_{fun} represent economic, safety, and driving function payoffs, respectively, the definitions of which are considered below.

Since a vehicle’s economy and comfort is related to the modulus of its acceleration vector, the economic functions used in this article are defined as in (2), where a_v denotes the acceleration of the single vehicle, a_i^p denotes the acceleration of vehicle i in the platoon, T_t indicates the current time, and T denotes the predicted period

$$\begin{cases} v_{\text{eco}} = \int_{T_t}^{T_t+T} a_v^2 dt \\ v_{\text{eco}}^p = \int_{T_t}^{T_t+T} a_p^2 dt \end{cases} \quad (2)$$

The safety payoff functions for single vehicles and platoons are evaluated below

$$\begin{cases} v_{\text{safe}} = (X_v - X_{p,b})^2 + (Y_v - Y_{p,b})^2 \\ = (X_v + V_{v,x}T - X_{p,b} - V_{v,x}T - 0.5a_{p,b}T^2)^2 \\ + (Y_v + V_{v,y}T + 0.5a_{v,y}T^2 - X_{p,b} - V_{v,x}T)^2 \\ v_{\text{safe}}^p = \sum_{i=1}^j (X_v - X_{p,i})^2 + (Y_v - Y_{p,i})^2 \end{cases} \quad (3)$$

Defining an expression for driving function payoff requires some assumptions about the cut-in behavior of the single vehicle. By abstracting this behavior as a problem of economic

benefits, we assume that the relevant function produces a positive return before completion, and a negative return after completion of cut-in, i.e., behavior that reduces lateral position tracking error is rewarded and behavior that increases lateral position tracking error is punished. Hence, with Y_{ref} defined as the reference lateral position of the vehicle, the path following function for defining cut-in behavior, is expressed as:

$$\begin{cases} v_{\text{fun}}^v = e_v^2 \\ v_{\text{fun}}^{p,i} = e_{p,i}^2 \end{cases} \quad (4)$$

Here, $e_v = Y_{\text{ref}} - Y$, $e_p = X_{\text{ref}} - X$ and denotes the single vehicle’s lateral following error, which converges to zero while $e\dot{e} < 0$, and $\dot{e}$ is the time derivative of e .

If both the platoon and the vehicles outside the platoon know each other’s payoff function and action set, the problem is a complete information game. Vehicles can make acceleration decisions for the purpose of maximizing their payoff function. When the two parties cannot share information perfectly, the conflicting parties can only make behavioral decisions by observing the actions of the other party. At this time, the problem is an incomplete information game.

III. ACCELERATION DECISIONAL METHOD BASED ON GAME THEORY

Assuming that merging vehicles and vehicles outside the platoon can obtain the cost function of each other through the communication or the cloud, the non-cooperative game scenario is a complete information dynamic game. Since the platoon actively accelerates and decelerates to give way, it will affect the efficiency and economy of the platoon, so it belongs to the passive responder and responds according to the actions of the merging vehicles.

Let the strategy of vehicles in the platoon be longitudinal acceleration $a_{p,i}$, and the strategy of merging smart cars as lateral acceleration $a_{v,y}$. Since acceleration is a continuous quantity in the decision-making process, a game with an infinite set of strategies is also called an infinite strategy game.

A. Existence Condition of Nash Equilibrium

With the given cost function in Section II, a dual-integrator dynamics model is used to predict the motion of the vehicles for expressing the explicit relationship between cost functions and action strategies. We may get the lateral motion equations of the cut-in vehicle and platoon with (6)

$$\begin{cases} X_v(t+T) = X_v(t) + V_{X,v}(t)T \\ Y_v(t+T) = Y_v(t) + V_{Y,v}(t)T + \frac{1}{2}a_{v,y}(t)T^2 \\ X_{p,i}(t+T) = X_{p,i}(t) + V_{p,i}(t)T + \frac{1}{2}a_{p,i}(t)T^2 \\ Y_{p,i}(t+T) = Y_{p,i}(t) + V_{p,i}(t)T \end{cases} \quad (5)$$

where T denotes the predicted time.

Subscribe (5) to the cost function. Then $v_v(a_v, a_{p,b})$ and $v_p(a_v, a_{p,1}, \dots, a_{p,j})$ will be obtained. The cost function of each vehicle takes the partial derivative of the strategy (acceleration) and sets it equal to 0, then the formula (6) can be obtained. There are $j+1$ equations and $j+1$ unknowns

in the formula: acceleration. Therefore, the Nash equilibrium solution can be obtained with the following equation:

$$\frac{\partial v_v}{\partial a_{v,y}} = 0, \frac{\partial v_{p,1}}{\partial a_{p,1}} = 0, \dots, \frac{\partial v_{p,j}}{\partial a_{p,j}} = 0. \quad (6)$$

The purpose of control is to minimize the cost function. Constrained by the saturation state of the actuator, the action strategies of the players in the game are bounded and need to satisfy $a \in [a_{\min}, a_{\max}]$. In summary, the sufficient condition to ensure that the game problem always has a Nash equilibrium solution is: the cost function is quasi-convex, and the cost function has a minimum value at this time.

While all participants act with the aim of cooperation, the cooperative cost function of the whole group is shown in the following equation:

$$J = v_v + \sum_{i=1}^j v_p^i. \quad (7)$$

At this time, the quasi-convexity of the cost function can be determined by the Hessian matrix, as shown in the following equation:

$$H = \begin{bmatrix} \frac{\partial^2 J}{\partial a_{v,y}^2} & \frac{\partial^2 J}{\partial a_{v,y} \partial a_{p,1}} & \dots & \frac{\partial^2 J}{\partial a_{v,y} \partial a_{p,j}} \\ \frac{\partial^2 J}{\partial a_{p,1} \partial a_{v,y}} & \frac{\partial^2 J}{\partial a_{p,1}^2} & \dots & \frac{\partial^2 J}{\partial a_{p,1} \partial a_{p,j}} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 J}{\partial a_{p,j} \partial a_{v,y}} & \frac{\partial^2 J}{\partial a_{p,j} \partial a_{p,2}} & \dots & \frac{\partial^2 J}{\partial a_{p,j}^2} \end{bmatrix}. \quad (8)$$

- 1) When the matrix H is a positive definite matrix, the cooperative cost function has a minimum value.
- 2) When the matrix H is a negative definite matrix, the cooperative cost function has a maximum value.
- 3) When the matrix H is a semi-positive definite matrix or a semi-negative definite matrix, there are “suspected” extreme points, which need to be determined by other methods.

There are three vehicles in platoon while $j = 3$. Hessian matrix can be calculated with the following equation, (9), as shown at the bottom of the next page.

Then, the sufficient conditions for the matrix Hessian to be positive definite and the cost function J to have a minimum value are shown in (10), at the bottom of the next page.

B. Acceleration Decision Method Based on Complete Information Dynamic Game

For CCF, the solution can be obtained by (11), as shown at the bottom of the next page.

It can be seen from the above formula that each equation has only one unknown, and the solution environment will not increase as the number of participants increases. And satisfy the linear relationship as shown in (12), at the bottom of the next page.

When the merging vehicle and the platoon are in the non-cooperative mode, the cost function is selfish [selfish cost function, (SCF)], and the equation relationship can be obtained by (13), as shown at the bottom of the next page.

Obviously, the difference between the profit function of cooperation and the profit function of selfishness reflects the difference between the two: whether the acceleration decision of the lane-changing vehicle considers the movement of the vehicles in the platoon.

IV. BEHAVIORAL DECISION METHOD BASED ON GAME THEORY OF INCOMPLETE INFORMATION

The control framework adopted in this article for the incomplete information game is shown in Fig. 2. To identify driving intentions, each party observes the driving states of different vehicles. Posterior probabilities for different vehicle types are then determined by combining these driving intentions with the prior probabilities in the belief pool. The decision tree and the probability of reaching a certain node are subsequently used to calculate the expected payoff.

A. Belief Pool

In incomplete information dynamic games, participants evaluate the expected return of an action using a belief pool based on assumed characteristics of preset types. In this study, type assignment is performed using the Harsanyi transformation, as shown in Fig. 3. Here, the original two-person incomplete information game becomes a three-person completely imperfect information game. The virtual participant's nature determines the platoon's type, forming two sub-games. O_1 and O_2 represent two prior probabilities, where $O_1 = 1 - O_2$. With this, the Bayesian Nash equilibrium can be used to find the current game's equilibrium solution. A belief pool should satisfy validity and consistency constraints. Here, validity is defined as the sum of the probabilities of encountering different vehicle types being equal to 1. For the platoon's belief pool to be consistent, vehicles in the platoon should share knowledge of the initial and updated states.

Since the participants do not know each other's specific characteristics, it is assumed that they can broadly be classified as strong types or weak types, as dictated by their behavior. The attributes associated with each type are summarized as follows.

Weak-Type Single Vehicle: The requirement for security has a dominant effect on the behavior of these types of vehicles. Therefore, although they have the will to cut in, if a collision is possible, they avoid changing lanes and choose another opportunity to do so.

Strong-Type Single Vehicle: These types of vehicles are more likely to attempt cut-in, as they suffer more loss of payoff if they are unable to complete this.

Weak-Type Platoon: The behavior of these types of platoons is dictated by safety and comfort benefits. Thus, they will tend to permit the cut-in of the single vehicle if the losses caused by deceleration are not large.

Strong-Type Platoon: This type of platoon operates in strict accordance with the preset spacing strategy. Hence, it tries to avoid the loss of driving efficiency and energy consumption caused by deceleration.

From the above definitions, it can be surmised that different vehicle types have different probabilities of actions. Hence, the

distribution of these probabilities for each vehicle type is listed in Table I. Information about these probabilities is assumed to be common prior knowledge. Thus, initially, the probability of action is determined by identifying the vehicle type, not the payoff function.

Where the a priori probabilities in the belief pool are updated using a combination of observations of historical actions and Bayesian estimation using equations of the form shown in (14). Here, the observed behavior of vehicles outside the platoon constitutes a system of driving events (such as B₁:

$$H = \begin{bmatrix} 2\alpha_v + 3\beta_p T^4 + \frac{1}{2}\beta_v T^4 + \frac{1}{2}\gamma_v T^4 & 0 & 0 & 0 \\ 0 & \frac{1}{2}\gamma_p T^4 + 2\alpha_p & \dots & 0 \\ 0 & 0 & \frac{1}{2}\gamma_p T^4 + 2\alpha_p & \vdots \\ 0 & 0 & 0 & \frac{1}{2}\gamma_p T^4 + 2\alpha_p \end{bmatrix} \quad (9)$$

$$\begin{cases} 2\alpha_v + 2\beta_v T^4 + \frac{1}{2}\gamma_v T^4 2\alpha_v + 2\beta_v T^4 + \frac{1}{2}\gamma_v T^4 > 0 \\ \frac{1}{2}\gamma_p T^4 + \alpha_p > 0 \end{cases} \quad (10)$$

$$\begin{aligned} \frac{\partial v}{\partial a_{v,y}} &= 2\alpha_v a_{v,y} + T^2 \beta_p \sum_{i=1}^3 \left(\frac{a_{v,y}}{2} T^2 + V_{v,y} T - Y_{p,i} + Y_v \right) + T^2 \beta_v \left(\frac{a_{v,y} T^2}{2} + V_{v,y} T - Y_{p,2} + Y_v \right) \\ &\quad + T^2 \gamma_v \left(\frac{a_{v,y} T^2}{2} + V_{v,y} T - Y_{v,\text{ref}} + Y_v \right) \\ \frac{\partial v_{p,1}}{\partial a_{p,1}} &= 2\alpha_p a_{p,1} + T^2 \beta_p \left(X_{p,1} - X_v + V_{p,x,1} T - V_{x,v} T + \frac{a_{p,1}}{2} T^2 \right) \\ &\quad + T^2 \gamma_p \left(\frac{a_{p,1}}{2} T^2 + V_{p,x,1} T + X_{p,1} - X_{p,1,\text{ref}} \right) \\ \frac{\partial v_{p,2}}{\partial a_{p,2}} &= 2\alpha_p a_{p,2} + T^2 \beta_p \left(X_{p,2} - X_v + V_{p,x,2} T - V_{x,v} + \frac{a_{p,2}}{2} T^2 \right) \\ &\quad + T^2 \gamma_p \left(\frac{a_{p,2}}{2} T^2 + V_{p,x,2} T + X_{p,2} - X_{p,2,\text{ref}} \right) + T^2 \beta_v \left(\frac{a_{p,2} T^2}{2} + X_{p,2} - X_v + V_{p,x,2} - V_{v,x} T \right) \\ \frac{\partial v_{p,3}}{\partial a_{p,3}} &= 2\alpha_p a_{p,3} + T^2 \beta_p \left(X_{p,3} - X_v + V_{p,x,3} T - V_{x,v} + \frac{a_{p,3}}{2} T^2 \right) \\ &\quad + T^2 \gamma_p \left(\frac{a_{p,3}}{2} T^2 + V_{p,x,3} T + X_{p,3} - X_{p,3,\text{ref}} \right) \end{aligned} \quad (11)$$

$$\begin{aligned} a_{v,y} &= - \frac{T^2 \beta_p \sum_{i=1}^3 (V_{v,y} T - Y_{p,i} + Y_v) + T^2 \beta_v (V_{v,y} T - Y_{p,2} + Y_v) + T^2 \gamma_v (V_{v,y} T - Y_{v,\text{ref}} + Y_v)}{2\alpha_v + \frac{3T^4}{2} \beta_p + \beta_v \frac{T^4}{2} + T^2 \gamma_v \frac{T^2}{2}} \\ a_{p,1} &= - \frac{T^2 \beta_p (X_{p,1} - X_v + V_{p,x,1} T - V_{x,v}) + T^2 \gamma_p (V_{p,x,1} T + X_{p,1} - X_{p,1,\text{ref}})}{2\alpha_p + \frac{T^4 \beta_p}{2} + \frac{T^4 \gamma_p}{2}} \\ a_{p,2} &= - \frac{(T^2 \beta_p + T^2 \beta_v) (X_{p,2} - X_v + V_{p,x,2} T - V_{x,v} T) + T^2 \gamma_p (V_{p,x,2} T + X_{p,2} - X_{p,2,\text{ref}})}{2\alpha_p + \frac{T^4 \beta_p}{2} + \frac{T^4 \gamma_p}{2} + \frac{\beta_v T^4}{2}} \\ &\vdots \\ a_{p,i} &= - \frac{T^2 \beta_p (X_{p,i} - X_v + V_{p,x,i} T - V_{x,v}) + T^2 \gamma_p (V_{p,x,i} T + X_{p,i} - X_{p,i,\text{ref}})}{2\alpha_p + \frac{T^4 \beta_p}{2} + \frac{T^4 \gamma_p}{2}} \end{aligned} \quad (12)$$

$$\begin{aligned} \frac{\partial v}{\partial a_{v,y}} &= 2\alpha_v + T^2 \beta_v \left(\frac{a_{v,y} T^2}{2} + V_{v,y} T - Y_{p,2} + Y_v \right) + T^2 \gamma_v \left(\frac{a_{v,y} T^2}{2} + V_{v,y} T - Y_{v,\text{ref}} + Y_v \right) \\ \frac{\partial v_{p,1}}{\partial a_{p,1}} &= 2\alpha_p a_{p,1} + T^2 \beta_p \left(\frac{a_{p,1} T^2}{2} + V_{p,x,1} T - T V_{v,x} + X_{p,1} - X_v \right) + T^2 \gamma_p \left(\frac{a_{p,1} T^2}{2} + V_{p,x,1} T + X_{p,1} - X_{p,1,\text{ref}} \right) \\ \frac{\partial v_{p,2}}{\partial a_{p,2}} &= 2\alpha_p a_{p,2} + T^2 \beta_p \left(\frac{a_{p,2} T^2}{2} + V_{p,x,2} T - T V_{v,x} + X_{p,2} - X_v \right) + T^2 \gamma_p \left(\frac{a_{p,2} T^2}{2} + V_{p,x,2} T + X_{p,2} - X_{p,2,\text{ref}} \right) \\ &\vdots \\ \frac{\partial v_{p,i}}{\partial a_{p,i}} &= 2\alpha_p a_{p,i} + T^2 \beta_p \left(\frac{a_{p,i} T^2}{2} + V_{p,x,i} T - T V_{v,x} + X_{p,i} - X_v \right) + T^2 \gamma_p \left(\frac{a_{p,i} T^2}{2} + V_{p,x,i} T + X_{p,i} - X_{p,i,\text{ref}} \right) \end{aligned} \quad (13)$$

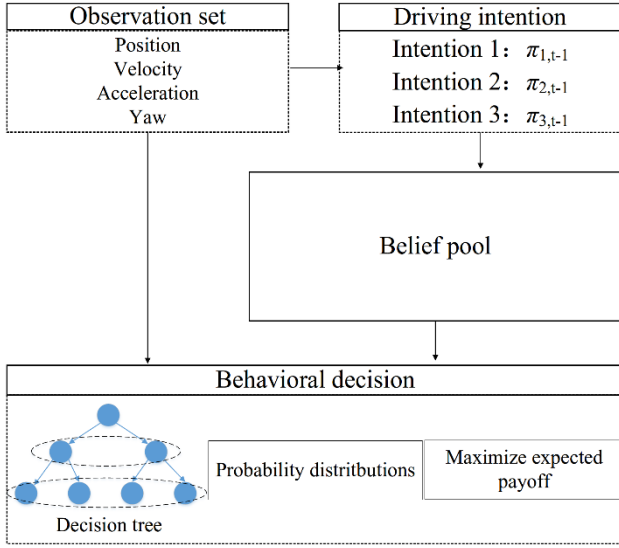


Fig. 2. Control flow for behavioral decision-making based on non-cooperative game theory.

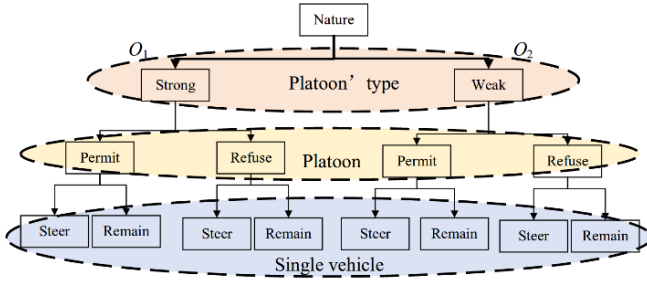


Fig. 3. Decision tree for the incomplete information dynamic game.

TABLE I

SUMMARY OF PROBABILITY DISTRIBUTIONS OF BEHAVIORS OF INTERACTING PARTIES

Probability distribution for single vehicle action			
Type	Probability	Steer	Remain
Strong	O_1	π_1	π_2
Weak	O_2	ψ_1	ψ_2
Sum	1	$\pi_1 \times O_1 + \psi_1 \times O_2$	$\pi_2 \times O_1 + \psi_2 \times O_2$
Probability distribution for platoon action			
Type	Probability	Permit	Refuse
Strong	Z_1	ϖ_1	ϖ_2
Weak	Z_2	Γ_1	Γ_2
Sum	1	$\varpi_1 \times Z_1 + \Gamma_1 \times Z_2$	$\varpi_2 \times Z_1 + \Gamma_2 \times Z_2$

steer to another lane or B_2 : remain in original lane), and the probability of occurrence is positive

$$P(A_m|B) = \frac{P(A_m)P(B|A_m)}{\sum_i P(A_i)P(B|A_i)}. \quad (14)$$

If it is observed that the vehicle has exhibited a specific behavior in the first game, the posterior probability is corrected as follows.

TABLE II
PAYOFF MATRIX

Vehicle \ Platoon	permit	refuse
remain	v_v^{11}, v_p^{11}	v_v^{12}, v_p^{12}
steer	v_v^{21}, v_p^{21}	v_v^{22}, v_p^{22}

For cut-in behavior, the posterior probability of the vehicle being strong is shown in (15), while the posterior probability of the vehicle being weak is shown in (16)

$$\tilde{O}_1 = P(A_1|B_1) = \frac{\pi_1 \times O_1}{\pi_1 \times O_1 + \psi_1 \times O_2} \quad (15)$$

$$\tilde{O}_2 = P(A_2|B_1) = \frac{\psi_1 \times O_2}{\pi_1 \times O_1 + \psi_1 \times O_2}. \quad (16)$$

For remain behavior, the posterior probability of the vehicle being strong is shown in (17), while the posterior probability of the vehicle being weak is shown in (18)

$$\tilde{O}_1 = P(A_1|B_2) = \frac{\pi_2 \times O_1}{\pi_2 \times O_1 + \psi_2 \times O_2} \quad (17)$$

$$\tilde{O}_2 = P(A_2|B_2) = \frac{\psi_2 \times O_2}{\pi_2 \times O_1 + \psi_2 \times O_2}. \quad (18)$$

The sum of the updated vehicle type distribution probabilities is still 1, as summarized below

$$\tilde{O}_1 + \tilde{O}_2 = \begin{cases} \frac{\pi_1 \times O_1 + \psi_1 \times O_2}{\pi_1 \times O_1 + \psi_1 \times O_2} = 1_{\text{steer}} \\ \frac{\pi_2 \times O_1 + \psi_2 \times O_2}{\pi_2 \times O_1 + \psi_2 \times O_2} = 1_{\text{remain}}. \end{cases} \quad (19)$$

The payoff matrix for different action set is shown in Table II.

Hence, the expected revenue when different action strategies are selected for vehicles and platoons can be obtained from the following calculation:

$$\begin{bmatrix} J_p^{\text{permit}} \\ J_p^{\text{refuse}} \\ J_v^{\text{remain}} \\ J_v^{\text{steer}} \end{bmatrix} = \Omega \begin{bmatrix} v_p^{11} \\ v_p^{21} \\ v_v^{11} \\ v_v^{21} \end{bmatrix} + \Phi \begin{bmatrix} v_p^{12} \\ v_p^{22} \\ v_v^{12} \\ v_v^{22} \end{bmatrix} \quad (20)$$

where J_p^{permit} and J_p^{refuse} , respectively, indicate the expected payoff when the platoon permits or rejects the cut-in, and J_v^{remain} and J_v^{steer} indicate the payoff for the single vehicle maintaining its original lane, and steering to another lane, respectively. Ω and Φ are the matrix descriptions of the previously-described behavior probability distributions, defined below, (21), as shown at the bottom of the next page.

Theorem: When the probability distribution in the belief pool satisfies the validity constraint, the trace of the $\Omega + \hat{O}$ matrix is equal to 4, as summarized below

$$\begin{aligned} \text{trace}(\Omega + \Phi) &= 2(\pi_1 O_1 + \psi_1 O_2 + \pi_2 O_1 + \psi_2 O_2) \\ &\quad + 2(\varpi_1 Z_1 + \Gamma_1 Z_2 + \varpi_2 Z_1 + \Gamma_2 Z_2) \\ &= 2(O_1 + O_2) + 2(Z_1 + Z_2) \\ &= 4. \end{aligned} \quad (22)$$

TABLE III
JOINT STRATEGY PROBABILITY DISTRIBUTION

Vehicle	Platoon	
	permit	refuse
remain	z	x
steer	c	l

When the expected payoffs for the platoons and single vehicles have been estimated, the decision set obtained by the Bayesian Nash equilibrium solution can be expressed as

$$\begin{aligned} (a_v^*, \sigma_v^*) &= \arg \max \{ \max [J_v^{\text{remain}}(a_v, \sigma_v, a_p, \sigma_v) \\ &\quad J_v^{\text{steer}}(a_v, \sigma_v, a_p, \sigma_p)] \} \\ (a_p^*, \sigma_p^*) &= \arg \max \{ \max [J_p^{\text{permit}}(a_v, \sigma_v, a_p, \sigma_v) \\ &\quad J_p^{\text{refuse}}(a_v, \sigma_v, a_p, \sigma_p)] \} \end{aligned} \quad (23)$$

where $a_v \in [a_v^{\min}, a_v^{\max}]$, $a_p \in [a_p^{\min}, a_p^{\max}]$, $\sigma_v \in \{0, 1, 2\}$, $\sigma_p \in \{0, 1, 2\}$.

The action distribution for finding the equilibrium solution of the current game is shown in Table III.

Here,

$$p = \begin{bmatrix} z(\text{per}, \text{ret}), x(\text{ref}, \text{ret}), \\ c(\text{per}, \text{ste}), l(\text{ref}, \text{ste}) \end{bmatrix}$$

denotes the relative probabilities for different action sets.

The relationship between probability distribution and vehicle types is shown in the following equation:

$$\begin{aligned} \begin{bmatrix} z \\ x \\ c \\ l \end{bmatrix} &= \begin{bmatrix} (\pi_1 O_1 + \psi_1 O_2) \times (\varpi_1 Z_1 + \Gamma_1 Z_2) \\ (\pi_1 O_1 + \psi_1 O_2) \times (\varpi_2 Z_1 + \Gamma_2 Z_2) \\ (\pi_2 O_1 + \psi_2 O_2) \times (\varpi_1 Z_1 + \Gamma_1 Z_2) \\ (\pi_2 O_1 + \psi_2 O_2) \times (\varpi_2 Z_1 + \Gamma_2 Z_2) \end{bmatrix} \\ z + x + c + l &= 1 \\ z, x, c, l &\geq 0. \end{aligned} \quad (24)$$

Assuming that the actual values of π, ψ, ϖ , and Γ are known, the relationship between joint strategy probability distribution and the probability in the belief pool is shown in Fig. 4. Through Fig. 4, we can find:

- 1) As the values of O_1 and Z_1 decrease, the probability of (remain, permit) will increase.
- 2) As O_1 decreases and Z_1 increases, the probability of (remain, refuse) will increase.
- 3) With the increase of O_1 and the decrease of Z_1 , the probability of (steer, permit) will increase.
- 4) With the increase of O_1 and Z_1 , the probability of (steer, refuse) will increase.

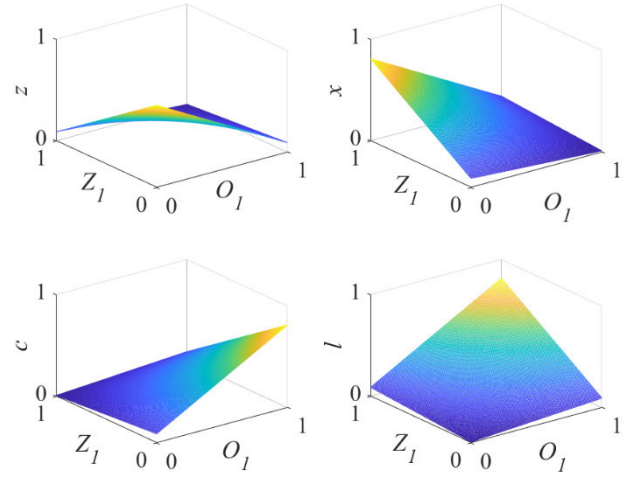


Fig. 4. Relationship between joint strategy probability distribution and the probability in belief pool.

B. Refined Bayesian Nash Equilibrium

At equilibrium, the inequality defined in (25) should hold

$$\begin{aligned} z v_v(\text{per}, \text{ret}) + x v_v(\text{ref}, \text{ret}) &\geq z v_v(\text{per}, \text{ste}) + x v_v(\text{ref}, \text{ste}) \\ c v_v(\text{per}, \text{ste}) + l v_v(\text{ref}, \text{ste}) &\geq c v_v(\text{per}, \text{ret}) + l v_v(\text{ref}, \text{ret}) \\ z v_p(\text{per}, \text{ret}) + c v_p(\text{per}, \text{ste}) &\geq z v_p(\text{ref}, \text{ret}) + c v_p(\text{ref}, \text{ste}) \\ x v_p(\text{ref}, \text{ret}) + l v_p(\text{ref}, \text{ste}) &\geq x v_p(\text{per}, \text{ret}) + l v_p(\text{per}, \text{ref}) \end{aligned} \quad (25)$$

where v_v denotes the payoff of the single vehicle and v_p denotes the payoff of the platoon.

Hence, by substituting the payoff matrix detailed in Table II to (25), the probability distribution shown in the following equation is obtained:

$$\begin{cases} z \geq \frac{v_v^{22} - v_v^{12}}{v_v^{11} - v_v^{21}} x \\ c \geq \frac{v_v^{12} - v_v^{22}}{v_v^{21} - v_v^{11}} l \\ z \geq \frac{v_p^{11} - v_p^{12}}{v_p^{21} - v_p^{22}} c \\ x \geq \frac{v_p^{21} - v_p^{22}}{v_p^{12} - v_p^{11}} l \end{cases} \quad (26)$$

According to the relationship shown in Fig. 4, we can get the relationship between the expected payoff of different actions and vehicle types with a certain matrix of payoff shown in Fig. 5. Surface 1 and 2 denote the expected payoff of different action (steer, remain). The values in surface are determined

$$\begin{aligned} \Omega &= \begin{bmatrix} \pi_1 O_1 + \psi_1 O_2 & 0 & 0 & 0 \\ 0 & \pi_1 O_1 + \psi_1 O_2 & 0 & 0 \\ 0 & 0 & \varpi_1 Z_1 + \Gamma_1 Z_2 & 0 \\ 0 & 0 & 0 & \varpi_1 Z_1 + \Gamma_1 Z_2 \end{bmatrix} \\ \Phi &= \begin{bmatrix} \pi_2 O_1 + \psi_2 O_2 & 0 & 0 & 0 \\ 0 & \pi_2 O_1 + \psi_2 O_2 & 0 & 0 \\ 0 & 0 & \varpi_2 Z_1 + \Gamma_2 Z_2 & 0 \\ 0 & 0 & 0 & \varpi_2 Z_1 + \Gamma_2 Z_2 \end{bmatrix} \end{aligned} \quad (21)$$

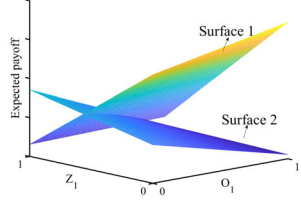


Fig. 5. Relation between expected payoff and the belief pool.

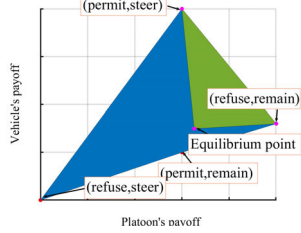


Fig. 6. Summary of possible payoff sets and Nash equilibrium solutions for the cut-in game.

TABLE IV

INITIAL STATE OF VEHICLES IN SIMULATION

Identity	Lane	Dynamics parameters			Driving intention	Behavioral distribution
		Position (m)	Velocity (m/s)	Acceleration (m/s ²)		
V_1	A	8	20	0	lane changing	(0.6, 0.4)
P_1	B	16	20	0	platooning	(0.5, 0.5)
P_2	B	8	20	0	platooning	(0.5, 0.5)
P_3	B	0	20	0	platooning	(0.5, 0.5)

by the belief pool, and when the value in Surface 1 is bigger than the value in surface 2, the action {steer} is dominant; otherwise, the action {remain} is dominant. It means that the results of behavioral decision are determined by the belief pool when the payoff of the action set is given.

Based on the above definitions, all game possible payoff sets and Nash equilibrium payoffs are plotted in Fig. 6. Here, the blue area represents all possible return points in the solution set, while the green area is the Nash equilibrium return, i.e., both sides meet Nash equilibrium when the expected return is in the green area.

V. EXPERIMENTS AND ANALYSIS

A. Experimental Setup

To verify the operation of our decision-making algorithm, we analyzed the interaction between different parties during a game simulation of cut-in. The objects of this game are a single vehicle in Lane A and a platoon of three vehicles in Lane B. The initial motion states and the key parameters for each party are summarized in Tables IV and V. The dynamic models for the cut-in vehicle and the platoon were built using CarSim, while the behavior decision-making algorithm and the motion controller for the vehicles were built in Simulink.

The co-simulation model's structure is illustrated in Fig. 7. The vehicle's and platoon's respective driving intentions are confirmed when they are initialized. Then, the driving intentions of other vehicles are determined based on the observed driving state information of surrounding targets. Both parties' respective belief pools are updated by combining the prob-

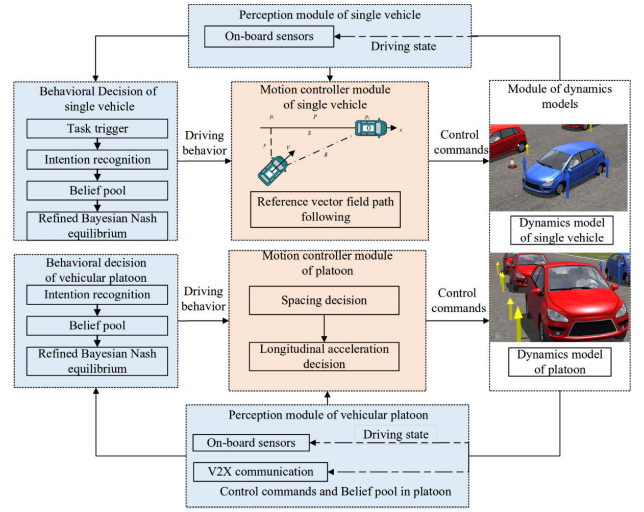


Fig. 7. Simulation framework for automated behavioral control of cut-in.

ability distribution function dictated by the relevant strategy with the Bayesian formula. This information is then used to predict the payoff for different behavioral types, and make a balanced behavioral decision. Acceleration decisions are subsequently made based on the estimated behavior, with inputs to the dynamic model regarded as control variables. In these simulations, the cut-in command for the intelligent vehicle in Lane A is triggered after 2 s, following which, the cut-in mode is turned on.

We also use a reference vector field (RVF) algorithm for lane-change motion control of the single vehicle [28]. Based on RVF, the lane-change trajectory that maintains a preset spacing with the preceding vehicle can be predicted as

$$y = y_0 e^{\frac{x-x_0}{S}} \quad (27)$$

where x and y denote the vehicle's longitudinal and lateral position, respectively, x_0 and y_0 indicate the vehicle's initial longitudinal and lateral position and S represents the spacing error.

In the proposed decision method, the parameters that affect the final acceleration include the prediction time period T in addition to the weight coefficient. To analyze the influence of different parameters on the merge result, this section simulates and compares the effects of the influencing parameters under different values, as shown in Fig. 8.

Through multiple simulations, the following conclusions can be drawn.

- 1) In Fig. 8, with the increase of α_v , the weight of comfort and economy increases, the vehicle will reduce the acceleration peak during the movement, and finally lead to the decrease of the error convergence speed of the lane change vehicle during the lane change process. As the acceleration decreases, the overshoot increases continuously during lateral movement, and the movement produces fluctuations.
- 2) It can be seen from Fig. 8(b) that the effect of β_v on the convergence speed of the tracking error of the lane-changing car is not significant enough. However, as the safety weight coefficient increases, the overshoot

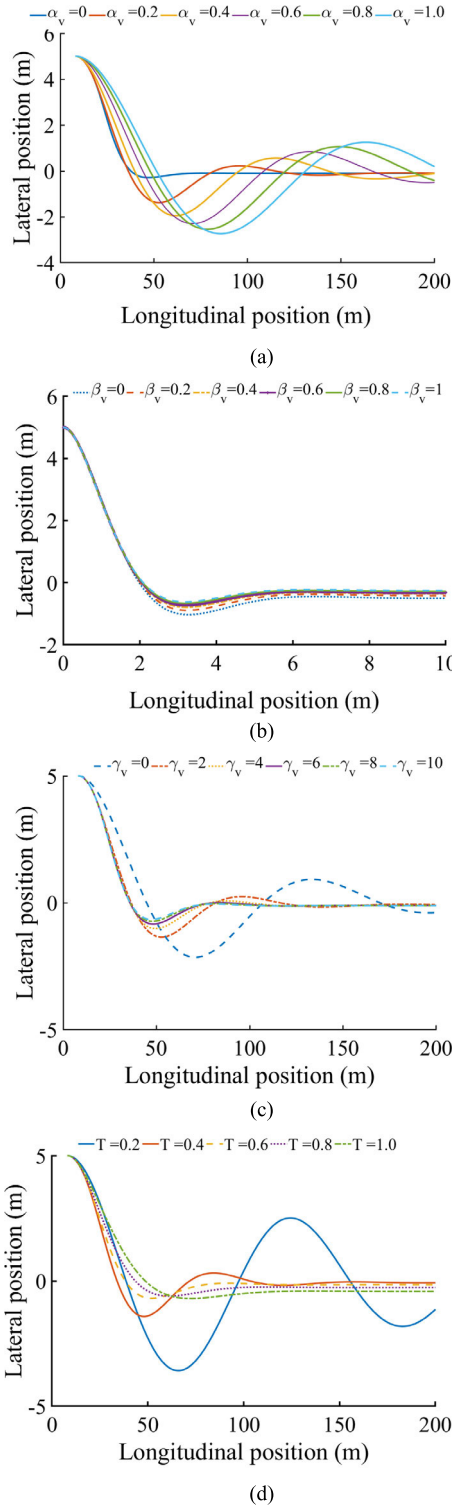


Fig. 8. Simulation results with different parameters. (a) α_v . (b) β_v . (c) γ_v . (d) T .

gradually increases. The reasons for this performance include two points: when deviating from the path, it will increase the relative distance with other vehicles and reduce the probability of collision with other vehicles; the increase of the safety weight coefficient reduces the influence of the path tracking weight on the vehicle movement.

TABLE V
DEFINITION OF CONTROL PARAMETERS USED IN SIMULATION

Symbol	Value	Symbol	Value
α_p	1	α_v	1
β_p	2	β_v	2
γ_p	1	γ_v	1
T	0.1	d	8

TABLE VI
TABLE OF NUMERICAL SIMULATION WEIGHT COEFFICIENT IN PROPOSED ACCELERATION DECISION METHOD

Name	Value
α_v	0.01
β_v	1
γ_v	5
α_p	0.01
β_p	1
γ_p	10
T	0.5

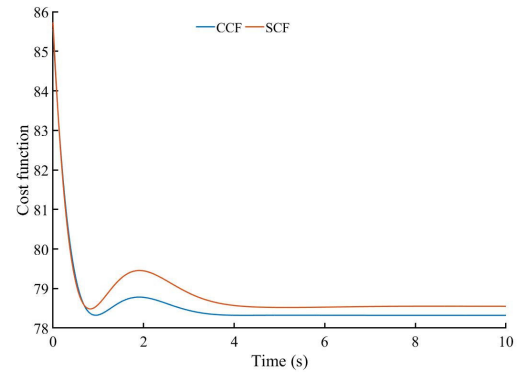


Fig. 9. Comparison of benefits of different cost functions.

- 3) In Fig. 8(c), with the increase of γ_v , the fluctuation of lateral tracking, overshoot, and error convergence speed are all improved. This is because the increase of the tracking function coefficient makes the vehicle gradually ignore factors such as safety and comfort, and the realization of the lane-changing function is becoming more and more important.
- 4) The forecast time T reflects the time window of profit calculation in each cycle. As the predicted time increases, the movement of the vehicle tends to be stable. The trajectory fluctuation caused by the adjustment difference of the weight coefficient is eliminated, but the error convergence speed gradually slows down.

The above analysis reflects the influence of the weight coefficient on lane-changing vehicles, in order to analyze the influence of the cooperative cost function and the selfish cost function on lane-changing vehicles and platoon. This article simulates lane-changing conditions with different cost functions under the same weight. The weight coefficients used in the simulation are shown in Table VI.

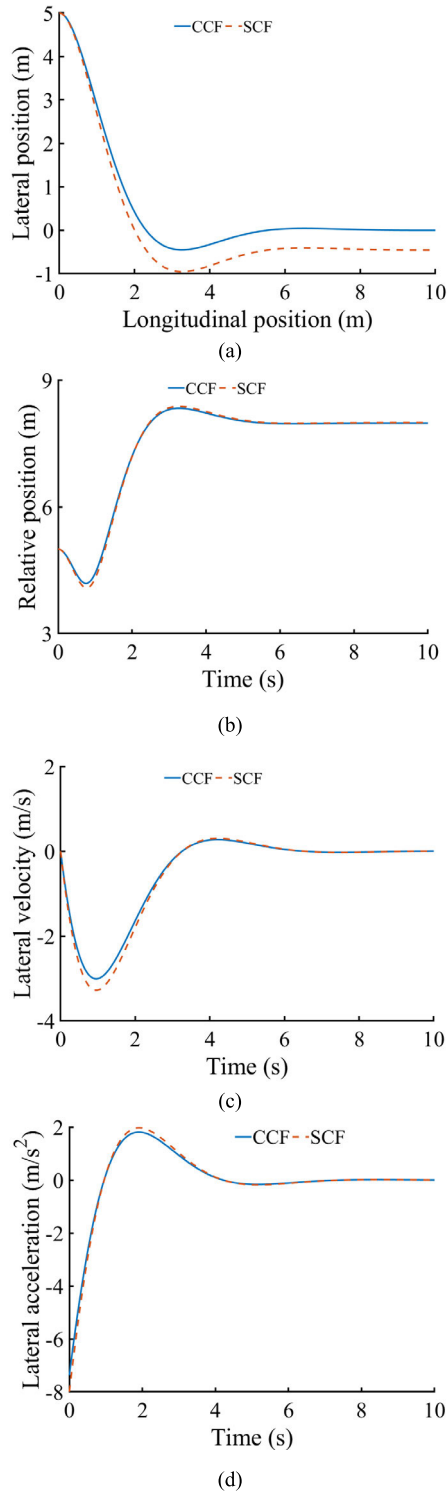


Fig. 10. Performance difference between CCF and SCF. (a) Path. (b) Relative distance between conflicted vehicle. (c) Lateral velocity. (d) Lateral acceleration.

Fig. 9 shows the cost-time curve of the merged smart car in CCF and SCF. In the entire time window, CCF is smaller than SCF, which shows that through CCF, the cut-in vehicle can pay less and achieve better control performance. This performance provides a motivation for cooperation between vehicle in this situation. From Fig. 10(a), it can be seen that when the final vehicle platoon finally reaches a stable state, the final static

error comes from the lateral tracking error. The determined lateral acceleration is shown in Fig. 10(d).

By comparing the cooperative cost function and the selfish cost function, we can find that, compared with SCF, CCF can effectively eliminate the lateral tracking error. The static error in SCF comes from the influence of other weights on the path tracking weight: when the vehicle continues to use acceleration to eliminate the lateral tracking error, it will increase the cost of comfort and safety. Fig. 10(b)–(d), respectively, shows the relative distance of the conflicting vehicle, the lateral speed of the merging vehicle, and the lateral acceleration. Among them, the vehicle under the control of CCF has smaller speed and acceleration peaks and is better than the performance of SCF in the minimum distance from the conflicting vehicle. Through CCF, the cut-in vehicle not only considers the revenue of conflicting vehicles but also considers the revenue of other vehicles in the platoon, thereby maximizing payoff.

In actual driving operations, smart cars mostly aim to maximize their own cost function. However, through the “V2V” and “V2C” technologies, comprehensive consideration of the cost function of vehicles in a certain area, and the formation of CCF for collaborative driving has potential research and application value.

B. Simulation Results of Behavioral Decision Method

In actual driving scenarios, the single vehicle and the platoon do not know each other’s detailed payoff function or specific behavioral type. Hence, in their own belief pool, the type probability for the interacting party is initially preset as $\{1/2, 1/2\}$. Fig. 11 shows curves for the belief pools of the intelligent vehicle and the platoon updated according to the behavior of the other party; the single vehicle has judged the platoon’s type as weak, and the platoon has judged the single vehicle’s type as strong. The convergence rate for the final type probability is related to the preset probability and the sampling period for the different behavior strategies.

To select an appropriate action, the single-vehicle and the platoon calculate the payoff for their adopted strategies according to their judgment of the other party’s type, and choose the behavior with the largest expected payoff. The payoffs of the vehicles during cut-in are shown in Fig. 12. For the single vehicle (V_1), the {steer} action always supersedes the {remain} action, since its expected payoff is larger. Similarly, since {refuse} always produces a better payoff than {permit} for vehicle P_1 in the platoon, it is the dominant action. In contrast, the {permit} action supersedes the {refuse} action for P_2 and P_3 .

The behavior decision resulting from the assumptions of the type of party and the expected payoff is shown in Fig. 13. Since V_1 has judged that the platoon is weak, it chooses a continuous cut-in strategy. In the belief pool update, to satisfy the validity constraint, the sum of the type probabilities should always be 1. While the platoon has judged V_1 as a strong type, each intelligent vehicle in the platoon is required to judge the vehicle type consistently. Hence, to satisfy the consistency principle, P_2 and P_3 , which conflict with V_1 , choose the same strategy. Consequently, the behavior of the interacting party

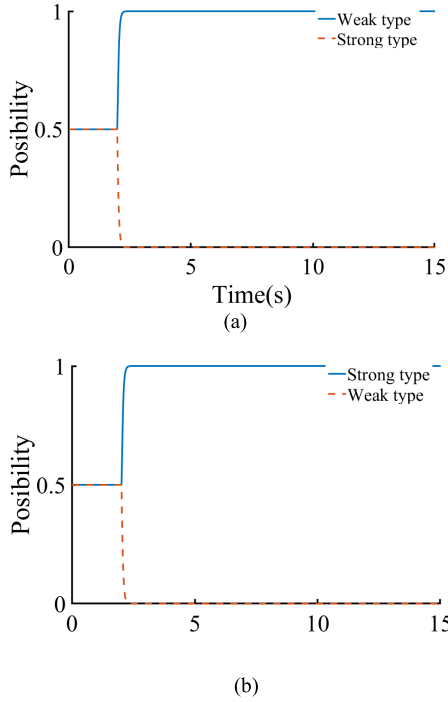


Fig. 11. Belief pool update during the cut-in process. (a) Belief pool of single vehicle judging platoon. (b) Belief pool of platoon judging the single vehicle.

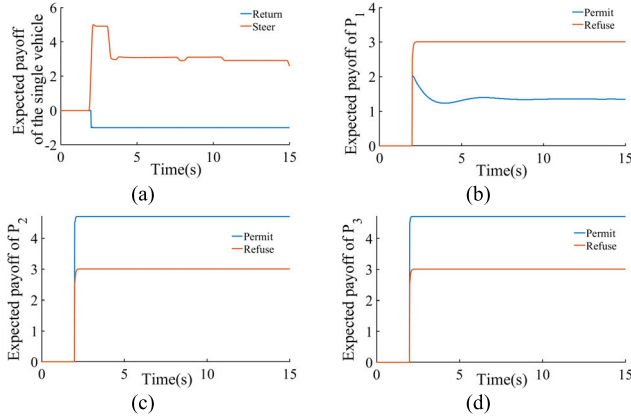


Fig. 12. Expected payoffs for the conflicting vehicles in simulation. (a) V_1 . (b) P_1 . (c) P_2 . (d) P_3 .

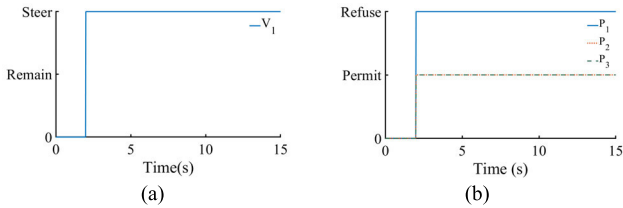


Fig. 13. Behaviorally-decided vehicle action. (a) V_1 . (b) Platoon.

conforms to the preset beliefs of each vehicle in the platoon, forming a stable game process. Therefore, in this game, the refined Bayesian Nash equilibrium solution is {steer, refuse, permit, permit}.

External vehicles enter the platoon by changing lanes, forming a new platoon, as shown in Figs. 14 and 15. P_2 and P_3 in the original platoon began to decelerate after 2 s, when cut-in was initiated, creating a space for V_1 to complete this

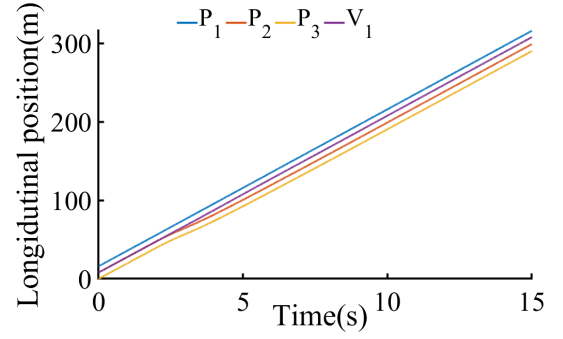


Fig. 14. Longitudinal position of the interacting vehicles.

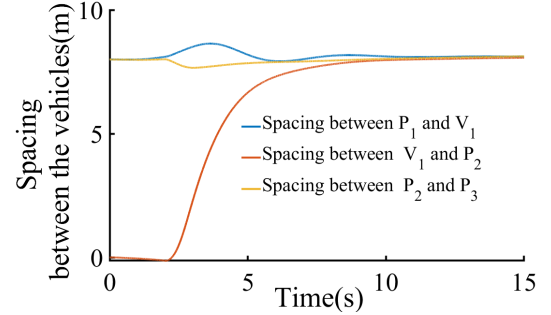


Fig. 15. Relative longitudinal position of the interacting vehicles.

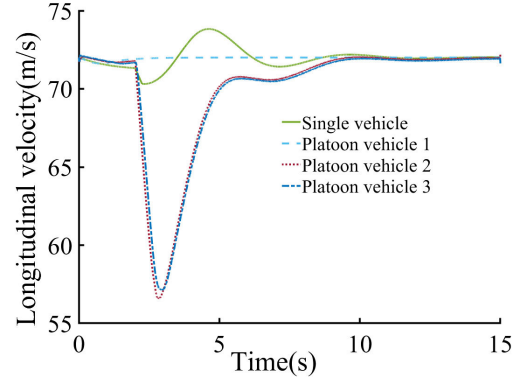


Fig. 16. Velocities of the interacting vehicles.

action. From these images, it can be seen that V_1 , P_2 , and P_3 return to a stable state at the same approximate time. This is because the agreement for resolving the conflict, to treat the single vehicle as a member of a new platoon, is reached relatively quickly. By following a distance control strategy in this platoon, all vehicles achieve coordinated longitudinal driving and enter a new stable state.

The parallel working conditions of the vehicles involved in the game are divided into four stages based on the relative spacing between conflicting vehicles (V_1 and P_2), as shown in Figs. 16 and 17. These stages are defined as follows:

a) *Stable Driving Stage Before Cut-In*: In this stage, the single vehicle and the original platoon are driving steadily along their current lanes. The single-vehicle decides to try the cut-in task at the end of this stage.

b) *Game Stage*: In this stage, the single-vehicle cuts in according to its belief pool and payoff function. Vehicles following in the platoon that conflict with the single-vehicle

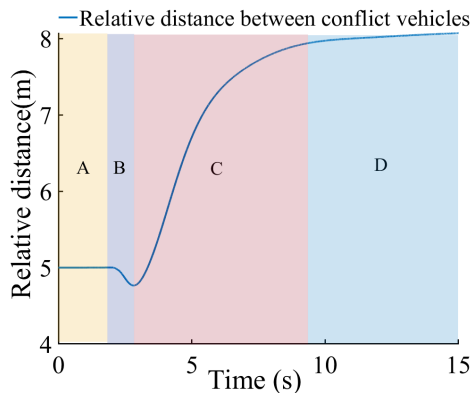


Fig. 17. Definition of stages involved in algorithm completion.

make decisions based on the platoon's shared belief pool and payoff function, deciding to give up space or prevent cut-in. The duration of this stage is related to the belief pool update function's frequency, and the payoff function. At the end of this stage, vehicles have reached a Bayesian Nash equilibrium {steer, permit}.

c) *Cooperative Cut-In Stage*: In this stage, the single vehicle is considered a member of a new vehicle platoon, and its formal status will be used as the basis for providing spacing control for the following vehicles.

d) *Stable Stage*: In this stage, vehicles in the new platoon enter a new stable driving state dictated by the spacing control strategy. The game process is over.

VI. CONCLUSION

In this study, we proposed a cooperative behavioral decision-making method for addressing road-rights conflict based on game theory, for solving driving situations with complete information and incomplete information. Two road-rights conflicting situations are analyzed and solved with different game models. The conclusions can be summarized as follows.

- 1) In situations with complete information, the lane-changing vehicle in CCF considers the influence of the other vehicles in platoon, hence the vehicle platoon with the obtained acceleration strategy can achieve a better synergy relationship, which is demonstrated to be better than in SCF. It shows that when sharing the cost function information can be realized through vehicle communication technology, the cooperative control based on the proposed non-cooperative game theory between multiple vehicles can achieve better performance.
- 2) Considering that the cost function information cannot be exchanged between vehicles, or the cost function information cannot be evaluated uniformly, the dynamic game model with incomplete information can update the belief pool according to the driving intention of the other vehicles and then select the behavior strategy with the best expected return to refine the Bayesian Nash equilibrium solution; in this equilibrium solution, neither party in the game has the motivation to change the current driving strategy or actively change the prior

information in the belief pool. Hence, in this case, the whole platoon system will be remaining stable condition based on the proposed non-cooperative game theory.

The significance of this study has been highlighted by noting the potential benefits for a more complex vehicle platoon driving scenarios rather than for solving the simple road-rights conflicting problems. The innovative point in this approach comes from the idea that conducting cooperative game between the vehicles inside the platoon and non-cooperative game between the external vehicles and the whole vehicle platoon. The mentioned conflicting problems will happen in several typical situations including cut-in, unmannered intersection, and roundabout.

The behavioral decision-making technique discussed in this article enables the possibility of fully automated platoon control, by providing a framework for addressing situations in which the intent of vehicles external to the platoon can affect its operation. While, in this study, complete information game theory and incomplete information game theory have been used for acceleration decision and behavioral decision-making, with further research, the principles of the technique can be adopted in continuous infinite strategy acceleration decision algorithms for cooperative control between vehicles. At this time, the payoff function for the action set should be quasi-concave, so that it has a calculable extremum. The parameters that characterizing the vehicle types can be the weight coefficient, however, the analysis of how these parameters affect the speed and the ability to reach an equilibrium will be studied in future.

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Yang Liu was born in Weifang, Shandong, China. He received the B.E. and M.S. degrees in vehicle engineering from the Shandong University of Science and Technology, Qingdao, China, in 2014 and 2017, respectively, and the Ph.D. degree in vehicle engineering from Jilin University, Changchun, China, in 2021.

His current research interests include multi-vehicles control and game theory.



Changfu Zong received the B.S. degree from the Liaoning University of Technology, Jinzhou, China, in 1986, and the M.Eng. and Ph.D. degrees from the Jilin University of Technology, Changchun, China, in 1994 and 1998, respectively.

In 2005, he joined Cambridge University, Cambridge, U.K., as an Academic Visitor. In 2013, he joined the University of California at Berkeley, Berkeley, CA, USA, as a Senior Academic Visitor. He is currently a Professor with the department, State Key Laboratory of Automotive Simulation and Control, Jilin University, Changchun. He has authored or coauthored over 200 articles. His current research interests include vehicle control stability, new energy vehicle, intelligent networked automobile control, and autonomous vehicle control.



Changhua Dai received the B.S. degree from Jilin University, Changchun, China, in 2014, where he is currently pursuing the Ph.D. degree with the College of Automotive Engineering.

His current research interests include automotive dynamics simulation and control.



Hongyu Zheng received the B.S. degree in mechanical engineering and automation and the Ph.D. degree in vehicle engineering from Jilin University, Changchun, China, in 2003 and 2009, respectively.

From October 2017 to October 2018, he was a Visiting Research Scholar with the Department of Mechanical and Aerospace Engineering, The Ohio State University, Columbus, OH, USA. He is currently a Professor with the State Key Laboratory of Automotive Simulation and Control, Jilin University.

His current research interests include vehicle dynamics and control and control of autonomous vehicles.



Dong Zhang (Member, IEEE) received the M.Sc. degree from Jilin University, Changchun, China, in 2015, and the Ph.D. degree from the University of Lincoln, Lincoln, U.K., in 2019.

In 2021, he joined Brunel having spent one year with Nanyang Technological University, Singapore, as a Research Fellow with the School of Mechanical and Aerospace Engineering and the School of Electrical and Electronic Engineering. He has been with Brunel and founded the Intelligent Driving and Transportation Research Group since September

2021. He is currently a Lecturer with the Department of Mechanical and Aerospace Engineering, Brunel University London, Uxbridge, U.K. He has authored or coauthored more than 30 articles in international journals as well as numerous conference articles, and about ten granted patents mostly in the area of intelligent driving and transportation control and active safety systems for road vehicles.